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Activity - 5

Sayyid Ahmad Alisha
CIS2541

- ① given Input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 }
and a hash function $h(x) = x \pmod{10}$, show the resulting
- separate chaining hash table
 - open addressing hash table using linear probing
 - open addressing hash table using quadratic probing
 - open addressing hash table with second with

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a) separate chaining:

$$h(x) = x \pmod{10}$$

$$h(4371) = (4371 \times 10) = 1$$

$$h(1323) = (1323 \times 10) = 3$$

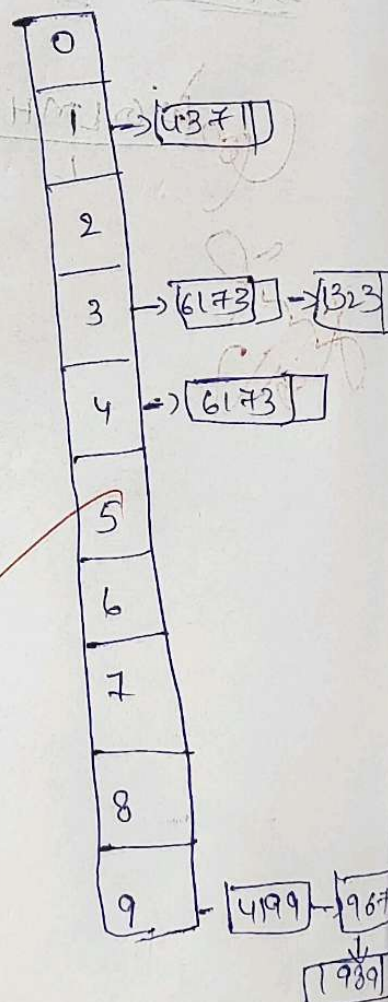
$$h(6173) = (6173 \times 10) = 3$$

$$h(4199) = (4199 \times 10) = 9$$

$$h(4344) = (4344 \times 10) = 4$$

$$h(9679) = (9679 \times 10) = 9$$

$$h(1989) = (1989 \times 10) = 9$$



b) linear hashing:

$$h(x) = h(x_{i-1} + 1)$$

$$h(4371) = 4371 \times 10 = 1$$

$$h(1323) = 1323 \times 10 = 3$$

$$h(6173) = 6173 \times 10 = 3$$

$$h(6173, i) = (6173 + 0) = 6173 \times 10 = 3$$

P=1 collision

n

$$(6173 + 1) = 6174 \times 10 = 4 \Rightarrow P=2$$

$$h(4199) = 4199 \times 10 = 9$$

$$h(4344) = 4344 \times 10 = 4$$

collision

P=1 collision

$$(4344 + 1) = 4345 \times 10 = 5 \Rightarrow P=2$$

$$h(9679) = 9679 \times 10 = 9$$

collision

$$h(9679, i) = (9679 + 0) = 9679 \times 10 = 9$$

P=1

$$9679 + 1 = 9680 \times 10 = 9$$

P=2

$$h(1989) = 1989 \times 10 = 9$$

collision

$$h(1989, i) = (1989 + 0) = 1989 \times 10 = 9$$

P=1

$$1989 + 1 = 1990 \times 10 = 0$$

P=2 collision

$$1989 + 2 = 1991 \times 10 = 1$$

P=3 collision

0	→ 9679
1	→ 4371
2	→ 1989
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	
9	→ 4199

$$1989 + 3 \Rightarrow 1992 \% 10 = 2$$

$$P = 4$$

c) Quadratic Hashing

$$h(key) = h(key + i^2)$$

↳ to tabulate

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

$$h(6173) = 6173 \% 10 = 3$$

collision

$$h(6173, i) = 6173 + i = 6174 \% 10 = 4$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4$$

collision

$$h(4344, i) = 4344 + i = 4345 \% 10 = 5$$

$$h(9679) = 9679 \% 10 = 9$$

collision

$$h(9679, i) = 9679 + i = 9680 \% 10 = 0$$

$$h(1989) = 1989 \% 10 = 9$$

collision

$$h(1989 + 4) = 1993 \% 10 = 3$$

$P = 2$ collision

$$h(1989 + 9) = 1998 \% 10 = 8$$

0	→ 9679
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	→ 1989
9	→ 4199

③ Double hashing

$$n(\text{key}) = [h_1(\text{key}) + h_2(\text{key}) * i]$$

$$h_1(\text{key}) = \text{key} \text{ mod } ts$$

$$h_2(\text{key}) = R - \text{key} \text{ mod } R$$

$$R = 1023$$

$$R \neq 7$$

$$h(4371) = 4371 \times 10 = 1$$

$$h(1323) = 1323 \times 10 = 3$$

$$h(6173) = 6173 \times 10 = 3$$

collision

$$h(6173) = 3 + [7 - 6] * 1$$

$$= 3 + 1$$

$$= 4$$

$$h(4199) = 4199 \times 10 = 9$$

$$h(4344) = 4344 \times 10 = 4$$

collision

$$h(4344) = 4 + [7 - 4] * 1$$

$$= 4 + 3 = 7$$

$$h(9679) = 9679 \times 10 = 9$$

collision

$$h(9679) = 9 + [7 - 9] * 1$$

$$= 9 + 2 = 11$$

collision

$$9 + [7 - 9] * 2$$

$$= 9 + 4 = 13$$

collision

$$9 + [7 - 9] * 3$$

$$= 9 + 6 = 15$$

0

1

2

3

4

5

6

7

8

9

→ 4371

→ 1323

→ 6173

→ 9679

→ 4344

→ 4199

$$n(1989) = 4 \cdot 1989 \times 10 = 9 \text{ collision}$$

$$n(1989) = [9 + (7-1) \times 1]$$

$$= 15 \text{ collision}$$

$$9 + [7-1] \times 2$$

$$= 9 + 12$$

$$= 21 \text{ collision}$$

$$9 + [7-1] \times 3$$

$$= 27 \text{ collision}$$

1989 does not have index position because
i. k-value has extended the tabular value

Good

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